

Practice Quiz: Perception / Electrophysiology

(Research Methods)

Name: __ Date: __

Part A: Multiple Choice

1. Mismatch negativity (MMN) indexes: A) Motor planning B) Sensory prediction error — the brain's response to stimuli that violate predictions, with amplitude reflecting PE magnitude C) Reward D) Sleep
2. Alpha oscillations (8-12 Hz) in Active Inference reflect: A) Motor commands B) Precision modulation — increased alpha suppresses irrelevant prediction errors, functionally implementing precision down-weighting C) Sensory input D) Random noise
3. Bottom-up prediction errors preferentially travel in: A) Alpha/beta bands B) Gamma and theta frequency bands, consistent with their role in driving fast, stimulus-related neural computation C) Delta only D) No specific band
4. The roving oddball paradigm tests: A) Motor learning B) Belief updating — how repeated exposure strengthens predictions, producing larger prediction errors when expectations are violated C) Emotional regulation D) Language processing
5. Laminar recordings can distinguish: A) Only noise B) Ascending prediction errors (superficial layers L2/3) from descending predictions (deep layers L5/6), directly testing the canonical microcircuit model C) Only one layer D) Chemical signals
6. The local-global paradigm dissociates: A) Two types of reward B) Local prediction errors (automatic, early, temporal cortex) from global prediction errors (awareness-dependent, late, frontal-parietal) — testing hierarchical prediction C) Vision from audition D) Memory from attention
7. Steady-state visual evoked potentials (SSVEPs) test Active Inference by: A) Measuring saccades B) Tagging neural responses to specific stimuli at known frequencies — allowing

researchers to track how attention (precision weighting) modulates the gain of specific sensory channels C) Testing motor reflexes D) Measuring sleep stages

8. Magnetoencephalography (MEG) advantages over EEG for testing Active Inference include: A) Better temporal resolution B) Superior spatial resolution and less sensitivity to volume conduction — enabling better localization of prediction error sources in specific cortical regions C) Cheaper equipment D) Better frequency range

Part B: Short Answer

1. A researcher records EEG during an auditory oddball task and finds that MMN amplitude decreases when participants are told the deviants are "unimportant." Interpret this in Active Inference terms: what computational variable is being modulated by the instruction, and how does this relate to the neural response? (200 words)

2. Explain why cross-frequency coupling (e.g., gamma amplitude nested within beta phase) is predicted by hierarchical predictive coding. What would disruption of this coupling (e.g., by ketamine) predict for perception? (200 words)

Part C: Essay Questions

1. Design a comprehensive EEG experiment testing three specific predictions of Active Inference about perception: (a) prediction errors are precision-weighted, (b) predictions travel top-down in beta frequencies, and (c) higher hierarchical levels generate later, more frontal responses. For each prediction, specify the paradigm, expected neural signature, and statistical test. (600 words)

2. Compare EEG, MEG, and fMRI as methods for testing Active Inference predictions. For each modality: strengths, limitations, what it can uniquely reveal, and what it cannot. Design an ideal multimodal study combining two or more modalities. (400 words)

3. Critically evaluate: Some researchers argue that MMN and related signals can be explained by simpler adaptation mechanisms (stimulus-specific adaptation, SSA) without invoking prediction. How would you distinguish the Active Inference account from the adaptation account experimentally? (400 words)